

Areas Related to Circles

Perimeter and area of a circle, how to find the arc length and area of a sector, how to find the area of a segment, and how to handle combinations of plane figures, with the exact NCERT formulae and worked examples.

Introduction

You have known the formulae for the circumference and area of a circle since earlier classes. In this chapter we use them to measure parts of a circle, a **sector** (a slice like a piece of pizza) and a **segment** (the cap cut off by a chord), and then combine circles with other shapes such as squares, rectangles and triangles to find the area of more complicated figures.

All the formulae in this chapter come from one simple idea: if a full circle corresponds to an angle of 360° at the centre, then a part that subtends an angle θ at the centre is just the fraction $\theta/360$ of the whole.

Perimeter and Area of a Circle

For a circle of radius r and diameter $d = 2r$:

Circle, Perimeter and Area

Circumference (perimeter) of a circle:

$$C = 2\pi r = \pi d$$

Area of a circle:

$$A = \pi r^2$$

Where π is taken as $22/7$ unless stated otherwise, or as 3.14 if the question says so.

These two formulae are the starting point. A semicircle, for example,, has an arc length of πr and, if you include the diameter, a perimeter of $\pi r + 2r$. Its area is half the full area, $\frac{1}{2}\pi r^2$. Similarly, a quarter-circle (quadrant) has area $\frac{1}{4}\pi r^2$.

NCERT also uses the idea of a circular path or ring. The area between two concentric circles (an annulus) with outer radius R and inner radius r is simply the difference of the two areas: $\pi R^2 - \pi r^2 = \pi(R^2 - r^2)$. The same subtraction idea is used throughout the chapter.

Worked Example

The radius of a circular park is 35 m. Find its circumference and area.

$$\text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 35 = 220 \text{ m}$$

$$\text{Area} = \pi r^2 = \frac{22}{7} \times 35 \times 35 = 3850 \text{ m}^2$$

Areas of Sectors

A **sector** is the region enclosed by two radii and the arc between them. If you cut a circle along two radii, each piece you get is a sector. The angle between the two radii, measured at the centre, is called the **angle of the sector** (θ).

Sector, Arc Length and Area

For a circle of radius r and a sector with central angle θ (in degrees):

$$\text{Length of the arc} = \frac{\theta}{360} \times 2\pi r$$

$$\text{Area of the sector} = \frac{\theta}{360} \times \pi r^2$$

$$\text{Also useful: Area of sector} = \frac{1}{2} \times (\text{arc length}) \times r$$

The logic is straightforward: the whole circle is 360° with circumference $2\pi r$ and area πr^2 . A sector that spans θ° is $\theta/360$ of the full circle, so its arc and area are the same fraction of the circumference and area.

A sector with $\theta < 180^\circ$ is called a **minor sector**, and the larger remaining part is the **major sector**. Their angles add to 360° , and their areas add to πr^2 . So once you know one, the other is simply the total minus that value.

Semicircle ($\theta = 180^\circ$) and **quadrant** ($\theta = 90^\circ$) are just special cases: put $\theta = 180^\circ$ or $\theta = 90^\circ$ into the formula and you get $\frac{1}{2}\pi r^2$ and $\frac{1}{4}\pi r^2$, as expected.

Worked Example

Find the area of a sector of a circle with radius 14 cm and central angle 60° .

$$\begin{aligned}\text{Area} &= \frac{\theta}{360} \times \pi r^2 = \frac{60}{360} \times \frac{22}{7} \times 14 \times 14 \\ &= \frac{1}{6} \times \frac{22}{7} \times 196 = \frac{1}{6} \times 616 = 102.67 \text{ cm}^2\end{aligned}$$

So the area is **$308/3 \text{ cm}^2$** (about **102.67 cm^2**).

Worked Example

Find the length of the arc of the same sector ($r = 14 \text{ cm}$, $\theta = 60^\circ$).

$$\begin{aligned}\text{Arc length} &= \frac{\theta}{360} \times 2\pi r = \frac{60}{360} \times 2 \times \frac{22}{7} \times 14 \\ &= \frac{1}{6} \times 88 = \frac{44}{3} \text{ cm} \approx 14.67 \text{ cm}\end{aligned}$$

Areas of Segments

A **segment** is the region between a chord and the corresponding arc. A chord divides the circle into two segments: the smaller one is the **minor segment** and the larger one is the **major segment**. The area of a segment is found by subtracting the area of a triangle from the area of its sector.

Segment, Area

For a segment with central angle θ and radius r :

$$\text{Area of minor segment} = \text{Area of sector OAB} - \text{Area of } \triangle OAB$$

$$\text{Area of major segment} = \text{Area of circle} - \text{Area of minor segment}$$

Where O is the centre and AB is the chord. For the triangle OAB with two sides r and included angle θ :

$$\text{Area of } \triangle OAB = \frac{1}{2} \times r^2 \times \sin \theta$$

In practice at Class 10 level, when θ is 60° , 90° or 120° , the triangle area is found using standard values: for $\theta = 60^\circ$, $\triangle OAB$ is equilateral with area $(\sqrt{3}/4)r^2$; for $\theta = 90^\circ$, $\triangle OAB$ is right-angled isosceles with area $\frac{1}{2}r^2$.

Think of it like this: the sector OAB is like a slice of pizza with a curved edge. If you cut off the triangular part OAB (the straight-sided slice), what remains, the curved cap, is exactly the segment.

NCERT expects you to handle a few standard angles. The sine values you need are:

$$\sin 60^\circ = \sqrt{3}/2 \quad \sin 90^\circ = 1 \quad \sin 120^\circ = \sqrt{3}/2$$

So for a 60° sector, area of $\Delta OAB = \frac{1}{2} r^2 \times \sqrt{3}/2 = (\sqrt{3}/4)r^2$, which matches the equilateral triangle formula. For 90° , it gives $\frac{1}{2} r^2$.

Worked Example

Find the area of the minor segment of a circle of radius 14 cm, when the central angle is 60° . (Use $\pi = 22/7$, $\sqrt{3} = 1.73$.)

Area of sector = $60/360 \times \pi r^2 = 1/6 \times 22/7 \times 14 \times 14 = 102.67 \text{ cm}^2$ (from earlier).

Area of ΔOAB : since $\theta = 60^\circ$ and $OA = OB = r$, the triangle is equilateral with side 14 cm.

$$\text{Area of } \Delta OAB = (\sqrt{3}/4) r^2 = (\sqrt{3}/4) \times 196 = 49\sqrt{3} \approx 49 \times 1.73 = 84.77 \text{ cm}^2$$

$$\text{Area of minor segment} = 102.67 - 84.77 = 17.90 \text{ cm}^2$$

$$\text{Area of major segment} = \text{Area of circle} - \text{Minor segment} = 616 - 17.90 = \mathbf{598.10 \text{ cm}^2}.$$

Worked Example

Find the area of the segment when the central angle is 90° and $r = 7$ cm. (Use $\pi = 22/7$.)

$$\text{Area of sector} = 90/360 \times \pi r^2 = \frac{1}{4} \times 22/7 \times 49 = 38.50 \text{ cm}^2$$

$$\text{Area of } \Delta OAB = \frac{1}{2} \times r^2 \times \sin 90^\circ = \frac{1}{2} \times 49 \times 1 = 24.50 \text{ cm}^2$$

$$\text{Area of minor segment} = 38.50 - 24.50 = \mathbf{14 \text{ cm}^2}$$

Areas of Combinations of Plane Figures

Many real-life figures are not a single shape but a combination, for example, a rectangular field with a semicircular flower bed, a window made of a rectangle and a

semicircle, or a running track made of a rectangle and two semicircles. To find such areas, NCERT follows a single strategy: add or subtract the areas you already know.

Combinations of Plane Figures

To find the area of a combined figure:

1. Break the figure into known parts (circles, sectors, segments, triangles, rectangles, squares).
2. Calculate each part separately using the formulae above.
3. Add the parts if they together make the whole, or subtract if a part is a hole or overlap.

There is no new formula to memorise here, the skill is in seeing which parts to add and which to subtract. Always draw a rough diagram and label the radii and angles before calculating.

Worked Example

A square park of side 20 m has a semicircular flower bed on each side on the outside. Find the total area of the four flower beds.

Each flower bed is a semicircle with diameter equal to the side of the square, so radius $r = 10$ m.

$$\text{Area of one semicircle} = \frac{1}{2} \pi r^2 = \frac{1}{2} \times \frac{22}{7} \times 10 \times 10 = \frac{1100}{7} \approx 157.14 \text{ m}^2$$

$$\text{Area of four semicircles} = 4 \times \frac{1100}{7} = \frac{4400}{7} \approx 628.57 \text{ m}^2$$

This is the same as the area of two full circles of radius 10 m: $2 \times \pi \times 100 = 200\pi \approx 628.57 \text{ m}^2$.

Worked Example

A design is made of a circle of radius 7 cm with a sector of angle 90° removed (cut out). Find the area of the remaining design.

$$\text{Area of circle} = \pi r^2 = 22/7 \times 49 = 154 \text{ cm}^2$$

$$\text{Area of removed sector} = 90/360 \times \pi r^2 = \frac{1}{4} \times 154 = 38.50 \text{ cm}^2$$

$$\text{Area of remaining design} = 154 - 38.50 = \mathbf{115.50 \text{ cm}^2}$$

The remaining part is the major sector with angle 270° , and indeed $270/360 \times 154 = 115.50 \text{ cm}^2$.

Another common pattern is the area swept by a pendulum or a wiper blade, this is simply a sector whose radius is the length of the blade and whose angle is the angle of the sweep. Similarly, the cost of fencing a circular path uses the circumference, while painting uses the area.

Summary

- **Circumference** of a circle of radius r : $C = 2\pi r = \pi d$; **Area**: $A = \pi r^2$.
- For a sector with central angle θ (in degrees): **arc length** $= \theta/360 \times 2\pi r$; **sector area** $= \theta/360 \times \pi r^2$. Also, sector area $= \frac{1}{2} \times \text{arc length} \times r$.
- **Minor sector** has $\theta < 180^\circ$, **major sector** has $\theta > 180^\circ$; their angles sum to 360° and areas sum to πr^2 .
- **Area of minor segment** = area of sector OAB – area of $\triangle OAB$, where area of $\triangle OAB = \frac{1}{2} r^2 \sin \theta$ (for 60° : $\sqrt{3}/4 r^2$; for 90° : $\frac{1}{2} r^2$).
- **Area of major segment** = area of circle – area of minor segment.
- For **combinations of plane figures**, break the figure into known shapes, calculate each area, and add or subtract as needed.
- Use $\pi = 22/7$ unless the question states otherwise.

Practice Questions

1. Find the circumference and area of a circle of diameter 14 cm. (Use $\pi = 22/7$.)

2. A sector of a circle of radius 21 cm subtends an angle of 120° at the centre. Find its arc length and area.
-
3. Find the area of the minor segment of a circle of radius 10 cm when the central angle is 90° . (Use $\pi = 3.14$.)
-
4. A chord of a circle of radius 14 cm subtends an angle of 60° at the centre. Find the area of the corresponding minor segment. (Use $\pi = 22/7$, $\sqrt{3} = 1.73$.)
-
5. A square of side 14 cm has a quadrant of radius 7 cm cut from each corner. Find the area of the remaining part of the square.
-
6. Two concentric circles have radii 14 cm and 7 cm. Find the area of the ring (annulus) between them.